LINGYU ZHU

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EDUCATION EXPERIENCE

City University of Hong Kong (CityUHK) **Hong Kong, China**
Computer Science, PhD. Advisor: Shiqi Wang. *2021.01 - 2024.12*

The Hong Kong University of Science and Technology (HKUST) **Hong Kong, China**
Mechanical Engineering, Master. Advisor: Kai Tang. *2018.09 - 2019.09*

Wuhan University of Technology (WUHT) **Wuhan, China**
Naval Architecture and Ocean Engineering, Bachelor. *2014.09 - 2018.06*

WORKING AND VISITING ACTIVITIES

- **City University of Hong Kong (CityUHK)**. Postdoc. Advisor: Shiqi Wang. Timeline: 2025.01 - 2025.11
- **Nanyang Technological University (NTU)**. Visiting Student. Advisor: Weisi Lin. Timeline: 2024.10 - 2024.12
- **Peking University Institute of Information Technology (PKU)**. Intern. Advisor: Zhao Wang. Timeline: 2023.06 - 2024.01
- **Peng Cheng Laboratory (PCL)**. Intern. Advisor: Siwei Ma. Timeline: 2021.10 - 2022.10
- **City University of Hong Kong (CityUHK)**. Research Assistant. Advisor: Shiqi Wang. Timeline: 2019.08 - 2021.01

RESEARCH INTERESTS

- Intelligent **Enhancement** Model in the open world (low-light, foggy, snowy, rainy, *etc*)
- Visual Collaborative **Compression** Towards Human-Machine Vision
- Quality **Assessment** of Visual Data Based on Large Multimodal Model
- Computational Photography
- Generative Model

PUBLICATIONS AND PRE-PRINT (CITATIONS $\geq$ 300, H-INDEX: 8)

[1] **Lingyu Zhu**, Xiangrui Zeng, Bolin Chen, Shiqi Wang: DualVisionCodec: High-Quality, Low-Bitrate Facial Compression Diffusion Rendering with Explicit Physics Modeling for Machine and Human Vision, **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted).**

[2] **Lingyu Zhu**, Xiangrui Zeng, Bolin Chen, Peilin Chen, Yung-Hui Li, Shiqi Wang: Leveraging Diffusion Knowledge for Generative Image Compression with Frequency-Aware Fractal Band Learning, **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted).**

[3] Xiangrui Zeng*, **Lingyu Zhu***, Peilin Chen, Shiqi Wang and Sam Kwong: WienerFlow: Wiener-Adaptive Flow Matching for Perception and Fidelity Trade-off in Low-light Image Enhancement, **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (*Co-first Author, Submitted).**

[4] Xiangrui Zeng*, **Lingyu Zhu***, Wenhan Yang, Leung Wing Ho Howard, Shiqi Wang and Sam Kwong: Low-Light Image Enhancement via Diffusion Models with Semantic Priors of Any Region, **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2025. (*Co-first Author, Submitted).**

[5] **Lingyu Zhu**, Wenhan Yang, Baoliang Chen, Hanwei Zhu, Zhangkai Ni, Qi Mao and Shiqi Wang: Unrolled Decomposed Unpaired Learning for Controllable Low-Light Video Enhancement, **European Conference on Computer Vision (ECCV), 2024. (Poster).**

[6] **Lingyu Zhu**, Binzhe Li, Riyu Lu, Peilin Chen, Qi Mao, Zhao Wang, Wenhan Yang and Shiqi Wang: Learned Image Compression for Both Humans and Machines via Dynamic Adaptation, **International Conference on Image Processing (ICIP), 2024. (Oral).**

[7] **Lingyu Zhu**, Wenhan Yang, Baoliang Chen, Hanwei Zhu, Xiandong Meng and Shiqi Wang: Temporally Consistent Enhancement of Low-light Videos via Spatial-Temporal Compatible Learning, **International Journal of Computer Vision (IJCV), 2024**.

[8] Baoliang Chen*, **Lingyu Zhu***, Hanwei Zhu, Wenhan Yang, Linqi Song and Shiqi Wang: Gap-Closing Matters: Perceptual Quality Evaluation and Optimization of Low-Light Image Enhancement, **IEEE Transactions on Multimedia (TMM), 2023. (*Co-first Author).**

[9] **Lingyu Zhu**, Wenhan Yang, Baoliang Chen, Fangbo Lu and Shiqi Wang: Enlightening Low-light Images with Dynamic Guidance for Context Enrichment, **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2022**.

[10] Runmin Zhang, Zhu Yu, Si-Yuan Cao, **Lingyu Zhu**, Guangyi Zhang, Xiaokai Bai, Hui-liang Shen: AGCNet: Boosting Multi-View Indoor 3D Object Detection via Adaptive Geometry and Context Aware Projection, **International Conference on Computer Vision (ICCV), 2025. (Poster).**

[11] Hanwei Zhu, Haoning Wu, Zicheng Zhang, **Lingyu Zhu**, Yixuan Li, Peilin Chen, Shiqi Wang and Chris Wei Zhou: AGCNet: VQualA 2025 Challenge on Visual Quality Comparison for Large Multimodal Models: Methods and Results, **International Conference on Computer Vision Workshops (ICCVW), 2025. (Organizers).**

[12] Hanwei Zhu, Baoliang Chen, **Lingyu Zhu**, Shiqi Wang and Weisi Lin: DeepDC: Deep Distance Correlation as a Perceptual Image Quality Evaluator. **IEEE Transactions on Image Processing (TIP), 2025.**

[13] Baoliang Chen, Hanwei Zhu, **Lingyu Zhu**, Shanshe Wang, Jingshan Pan and Shiqi Wang: Debiased Mapping for Full-Reference Image Quality Assessment. **IEEE Transactions on Multimedia (TMM), 2025**.

[14] Bolin Chen, Shanzhi Yin, Hanwei Zhu, **Lingyu Zhu**, Zihan Zhang, Jie Chen, Ru-Ling Liao, Shiqi Wang and Yan Ye: Compressing Human Body Video with Interactive Semantics: A Generative Approach. **International Conference on Image Processing (ICIP), 2025. (Poster).**

[15] Bolin Chen, Shanzhi Yin, Zihan Zhang, Jie Chen, Ru-Ling Liao, **Lingyu Zhu** and Shiqi Wang, Yan Ye: Beyond GFVC: A Progressive Face Video Compression Framework with Adaptive Visual Tokens. **Data Compression Conference (DCC), 2025. (Oral).**

[16] Hao Luo, **Lingyu Zhu**, Yudong Mao, Yixuan Li, Zhiwei Zhong and Shiqi Wang: Exploiting Long and Short Temporal Dependence for Low-Light Video Enhancement. **IEEE International Conference on Multimedia & Expo (ICME), 2025. (Poster).**

[17] Hao Luo, Zhiwei Zhong, **Lingyu Zhu**, Yudong Mao and Shiqi Wang: MS-MoE: Multi-modal Structural Mixture of Experts Framework for Pan-Sharpening. **International Joint Conference on Neural Networks (IJCNN), 2025. (Poster).**

[18] Hao Luo, Baoliang Chen, **Lingyu Zhu**, Peilin Chen and Shiqi Wang: RCNet: Deep Recurrent Collaborative Network for Multi-View Low-Light Image Enhancement. **IEEE Transactions on Multimedia (TMM), 2024**.

[19] Hanwei Zhu, Haoning Wu, Yixuan Li, Zicheng Zhang, Baoliang Chen, **Lingyu Zhu**, Yuming Fang, Guangtao Zhai, Weisi Lin and Shiqi Wang: Adaptive Image Quality Assessment via Teaching Large Multimodal Model to Compare. **Neural Information Processing Systems (NeurIPS), 2024. (Spotlight).**

[20] Hanwei Zhu, Baoliang Chen, **Lingyu Zhu**, Peilin Chen, Linqi Song and Shiqi Wang: Video Quality Assessment for Spatio-Temporal Resolution Adaptive Coding. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2024**.

[21] Baoliang Chen, Hanwei Zhu, **Lingyu Zhu**, Shiqi Wang and Sam Kwong: Deep Feature Statistics Mapping for Generalized Screen Content Image Quality Assessment. **IEEE Transactions on Image Processing (TIP), 2024. (Second Prize, Guangdong Computer Academy).**

[22] Riyu Lu, **Lingyu Zhu**, Baoliang Chen, Xiaopeng Fan and Shiqi Wang: Diffusion-based Bit-depth Expansion. **International Workshop on Multimedia Signal Processing (MMSP), 2024. (Oral).**

[23] Hanwei Zhu, Baoliang Chen, **Lingyu Zhu**, and Shiqi Wang: Learning Spatiotemporal Interactions for User-generated Video Quality Assessment. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2022**.

[24] Baoliang Chen, **Lingyu Zhu**, Chenqi Kong, Hanwei Zhu, Shiqi Wang and Zhu Li: No-reference Image Quality Assessment by Hallucinating Pristine Features. **IEEE Transactions on Image Processing (TIP), 2022**.

[25] Liangwei Yu, Zhao Wang, Yan Ye, **Lingyu Zhu** and Shiqi Wang: A Soft-ranked Index Fusion Framework with Saliency Weighting for Image Quality Assessment. **The IEEE / CVF Computer Vision and Pattern Recognition Conference Workshop (CVPRW), 2022. (Second Place, 5th Workshop and Challenge on Learned Image Compression)**.

[26] Baoliang Chen, **Lingyu Zhu**, Guo Li, Hongfei Fan and Shiqi Wang: Learning Generalized Spatial-temporal Deep Feature Representation for No-reference Video Quality Assessment. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2021**.

[27] Guo Li, Baoliang Chen, **Lingyu Zhu**, Qingwen He, Hongfei Fan and Shiqi Wang: PUGCQ: A Large Scale Dataset for Quality Assessment of Professional User-generated Content. **ACM Multimedia (ACM MM), 2021. (Poster)**.

[28] Qianyu Zhang, Bolun Zheng, Hangjia Pan, **Lingyu Zhu**, Zunjie Zhu, Zongpeng Li, and Shiqi Wang: Capturing Stable HDR Videos Using a Dual-Camera System. **IEEE Transactions on Image Processing (TIP), 2025. (Submitted).**

[29] Songchao Tan, **Lingyu Zhu**, Peilin Chen, Shiqi Wang, Ruiqi Li, Siwei Ma and Huimin Ma: Quality Assessment for Machine Vision: A Review and Perspectives. **IEEE Multimedia (MM), 2025. (Submitted).**

[30] Jiaxi Huang, Dongxu Wu, Hanwei Zhu, **Lingyu Zhu**, Jun Xing, Xu Wang and Baoliang Chen: Q-Doc: Benchmarking Document Image Quality Assessment Capabilities in Multi-modal Large Language Models. **Chinese Conference on Pattern Recognition and Computer Vision (PRCV), 2025. (Submitted).**

[31] Zhihao Lin, Zhenshan Shi, Sasa Zhao, Hanwei Zhu, **Lingyu Zhu**, Baoliang Chen and Lei Mo: Simple Lines, Big Ideas: Towards Interpretable Assessment of Human Creativity from Drawings. **Chinese Conference on Pattern Recognition and Computer Vision (PRCV), 2025. (Submitted).**

[32] Zhicheng Liao, Dongxu Wu, Zhenshan Shi, Sijie Mai, Yuncheng Jiang, Hanwei Zhu, **Lingyu Zhu** and Baoliang Chen: Beyond Cosine Similarity: Magnitude-Aware CLIP for No-Reference Image Quality Assessment. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted)**

[33] Baoliang Chen, Qing Lin, Zhihao Lin, Zhicheng Liao, Sijie Mai, Hanwei Zhu and **Lingyu Zhu**: Bridging Adversarial and Collaborative Learning for Quality Assessment of AI-Generated Images. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted)**

[34] Baoliang Chen, Xinlong Bu, Zhihao Lin, Yuncheng Jiang, Hanwei Zhu and **Lingyu Zhu**: EduVQA: Benchmarking AIGC Video Quality Assessment for Education. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted)**

[35] Baoliang Chen, Siyi Pan, Dongxu Wu, Hanwei Zhu, **Lingyu Zhu** and Xiangjie Sui: Mitigating Perception Bias: A Training-Free Approach to Enhance LMM for Image Quality Assessment. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted)**

[36] Baoliang Chen, Zhihao Lin, Zhenshan Shi, Hanwei Zhu and **Lingyu Zhu**: Ideas in Ink: Human Drawing Creativity Assessment via CLIP-powered Feature Disentanglement. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted)**

[37] Yudong Mao, Peilin CHEN, Hao Luo, **Lingyu Zhu**, Yung-Hui Li and Shiqi Wang: Optimizing Fidelity-Perception Trade-Off via Large Vision-Language Model Prior for Image Compression. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted).**

[38] Wei Lu, **Lingyu Zhu** and Si-Bao Chen: U3D: An Unsupervised UAV Dataset and Benchmark for Ultra-High-Resolution Low-Light Image Enhancement. **Annual AAAI Conference on Artificial Intelligence (AAAI), 2026. (Submitted).**

[39] Bolin Chen, Hanwei Zhu, Shanzhi Yin, **Lingyu Zhu**, Jie Chen, Ru-Ling Liao, Shiqi Wang and Yan Ye: Pleno-Generation: A Scalable Generative Face Video Compression Framework with Bandwidth Intelligence. **IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025. (Submitted).**

PROFESSIONAL SERVICES

Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence **(TPAMI)**
- International Journal of Computer Vision **(IJCV)**
- IEEE Transactions on Image Processing **(TIP)**
- IEEE Transactions on Circuits and Systems for Video Technology **(TCSVT)**
- IEEE Transactions on Multimedia **(TMM)**
- IEEE Multimedia **(MM)**
- Journal of Visual Communication and Image Representation **(JVCI)**
- International Journal of Machine Learning and Cybernetics **(JMLC)**
- Computer Vision and Image Understanding **(CVIU)**
- Pattern Recognition **(PR)**
- Imaging Science Journal
- Signal, Image and Video Processing
- Machine Vision and Applications
- Discover Artificial Intelligence
- Cluster Computing
- The Visual Computer
- Multimedia Systems
- Scientific Reports

Conference Reviewer

- ACM International Conference on Multimedia **(ACM MM)**
- IEEE / CVF Computer Vision and Pattern Recognition Conference **(CVPR)**
- International Conference on Computer Vision **(ICCV)**
- European Conference on Computer Vision **(ECCV)**
- International Conference on Machine Learning **(ICML)**
- IEEE International Conference on Acoustics, Speech, and Signal Processing **(ICASSP)**
- IEEE International Workshop on Multimedia Signal Processing **(MMSP)**
- IEEE International Symposium on Circuits and Systems **(ISCAS)**
- IEEE International Conference on Multimedia & Expo **(ICME)**
- International Joint Conference on Neural Networks **(IJCNN)**
- Pattern Recognition and Computer Vision **(PRCV)**
- Picture Coding Symposium **(PCS)**

Organization Committee

- VQualA Workshop and Competition @ **(ICCV 2025 Workshops)**

MISCELLANEOUS

- Programming Skills: Python, Pytorch, Tensorflow, Matlab, C

ACADEMIC COMMUNITY

- IEEE Member. Timeline: 2023.12 - Now
- CSIG Member. Timeline: 2022.12 - Now

TEACHING ASSISTANT

- Semester A, CS3343, Software Engineering Practice, City University of Hong Kong. Timeline: 2024.09 - 2024.12
- Semester B, CS3342, Software Design, City University of Hong Kong. Timeline: 2024.01 - 2024.05
- Semester A, CS3343, Software Engineering Practice, City University of Hong Kong. Timeline: 2023.09 - 2023.12
- Semester B, CS3342, Software Design, City University of Hong Kong. Timeline: 2023.01 - 2023.05
- Semester A, CS3343, Software Engineering Practice, City University of Hong Kong. Timeline: 2022.09 - 2022.12
- Semester B, CS3342, Software Design, City University of Hong Kong. Timeline: 2022.01 - 2022.05
- Semester A, CS5481, Data Engineering, City University of Hong Kong. Timeline: 2021.09 - 2021.12
- Semester B, CS1102, Introduction to Computer Studies, City University of Hong Kong. Timeline: 2021.01 - 2021.05

HONORS AND AWARDS

- Second Prize, Guangdong Computer Academy. Timeline: 2024.12
- Outstanding Academic Performance Award. Timeline: 2024.08 - 2025.08
- Conference Grant. Timeline: 2024.09 - 2024.10
- Institutional Research Tuition Scholarship. Timeline: 2023.09 - 2024.09
- Workshop and Challenge on Learned Image Compression: 2nd Place in the Perceptual Metric Track. Timeline: 2022.06
- Research Activities Fund. Timeline: 2021.10 - 2022.04
- Research Activities Fund. Timeline: 2021.01 - 2024.12
- Wuhan University of Technology CSC Scholarship. Timeline: 2014.11 - 2015.11